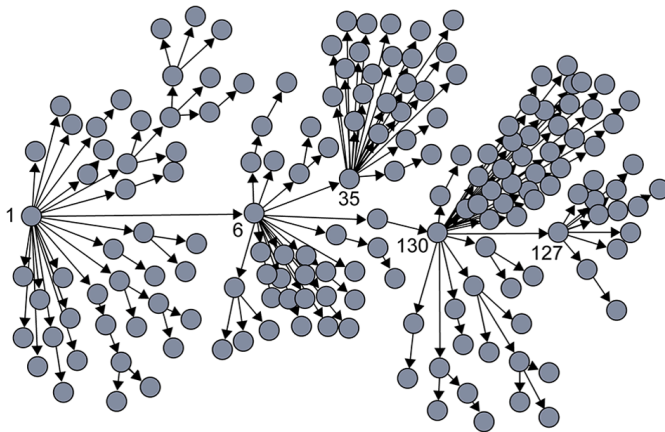


L4: Case Studies: Superspreaders

Superspreaders in SARS epidemic



The SARS (*Severa Acute Respiratory Syndrome*) was an epidemic back in 2002-3. It infected 8000 people in 23 countries and it caused about 800 deaths.

The plot shown here shows how the infections progressed from a single individual (*labeled as patient-1*) to many others. Such plots result from a process known as “contact tracing” – finding out the chain of successive infections in a population.

It is important to note the presence of a few hub nodes, referred to as “**superspreaders**” in the context of epidemics. The superspreaders are labeled with an integer identifier in this plot. The superspreader 1, for example, infected directly about 20

individuals.

The presence of superspreaders emphasizes the role of degree heterogeneity in network phenomena such as epidemics. If the infection network was more “**Poisson-like**”, it would not have superspreaders and the total number of infected individuals would be considerably smaller.

Source: Super-spreaders in infectious diseases [Richard A. Stein](https://www.sciencedirect.com/science/article/pii/S1201971211000245) [↗](https://www.sciencedirect.com/science/article/pii/S1201971211000245) (<https://www.sciencedirect.com/science/article/pii/S1201971211000245>), International Journal of Infectious Diseases, August 2011 <https://doi.org/10.1016/j.ijid.2010.06.020> [↗](https://doi.org/10.1016/j.ijid.2010.06.020) (<https://doi.org/10.1016/j.ijid.2010.06.020>)

Superspreaders Versus The Average Reproductive Number R_0

Disease	Location (year)	(R_0) ^a	SSE ^b	References
EBOLA	Congo (1995)	1.83	21+, 28–38	[85,86]
MEASLES	Greenland (1951)	16	250	[15,87]
	US (1985)	16	69,84	[15,88]
	Canada (2011)	16	678	[15,17]
PNEUMONIC PLAGUE	China (1946)	1.3	32	[89,90]
SARS	Hong Kong (2003)	3	187	[91,92]
	Vietnam (2003)	3	20	[91,93]
	Singapore (2003)	1.6	12,21,23,23,40+	[14,19]
	Canada (2003)	3	19,12–24	[91,94]
SMALLPOX	Yugoslavia (1975)	5.5	38	[95,96]

^a R_0 : The average number of secondary cases caused by an infected individual during the outbreak; here, R_0 is reported either for a specific outbreak, when available, or as a measure calculated based on multiple past outbreaks.

^bSSE: Superspreading events—number of infections caused by a single individual during an outbreak; number of infections caused by multiple superspreading events during the same outbreak are separated by commas.

This table is adapted from the supplementary material from reference [14].

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Epidemiologists often use the basic “**reproductive number**”, R_0 , which describes the average number of secondary infections that arise from one infected individual in an otherwise totally susceptible population.

One way to estimate R_0 is to multiply the average number of contacts of an infected individual by the probability that a susceptible individual will become infected by a single infected individual (“**shedding potential**”). So, the R_0 metric does not depend only on the given pathogen – it also depends on the number of contacts that each individual has. If $R_0 > 1$ then an outbreak is likely to become an epidemic, while if $R_0 < 1$ then an outbreak will not spread beyond a few initially infected individuals

It is important to realize however that R_0 is only an average – it does not capture the heterogeneity in the number of contacts of different individuals (*and it also does not capture the heterogeneity in the shedding potential of the pathogen at different individuals*). As we know by now, contact networks can be extremely heterogeneous in terms of the degree distribution, and they can be modeled with a power-law distribution of (*theoretically*) infinite variance. Such networks include hubs – and as we saw above, hubs can act as superspreaders during epidemic outbreaks.

The table in this page confirms this point for several epidemics. The third column shows R_0 while the fourth column shows **“Superspreading events”** (SSE). These are events during an outbreak in which a single infected individual causes a large number of direct or indirect infections. For example, in the case of the 2003 SARS epidemic in Hong Kong, even though R_0 was only 3, there was an SSE in which an infected individual caused a total of 187 infections (*patient-1 above*).

SSEs have been observed in practically every epidemic – and they have major consequences both in terms of the speed through which an epidemic spreads and in terms of appropriate interventions. For example, in the case of respiratory infections (*such as COVID-19*) **“social distancing”** is an effective intervention only as long as it is adopted widely enough to also include superspreaders.

Source: Cellular Superspreaders: An Epidemiological Perspective on HIV Infection inside the Body Kristina Talbert-Slagle et al., 2014, <https://doi.org/10.1371/journal.ppat.1004092> ⇨ (<https://doi.org/10.1371/journal.ppat.1004092>)